ME 144L LE 5: Beam Vibration and Accelerometer Usage

Alejandro Diaz
TA: Gabrielle Naquila

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1 Zero-G Measurements

To calibrate the accelerometer and gyro, the mean and standard deviation of 3 accelerometer axes and 3 gyroscope axes were computed from a near-stationary segment of the data (the final 2 seconds of the 20-second recording, after beam vibrations had fully decayed). The sensor was configured vertically upward, with the $\pm 8\,\mathrm{g}$ full-scale accelerometer range, which has a sensitivity $= 0.244\,\mathrm{mg/LSB}$, and the $\pm 125\,\mathrm{dps}$ gyroscope range sensitivity of $= 4.375\,\mathrm{mdps/LSB}$.

Channel	**Mean**	**Std. Dev**
Acc X	0.069 g	0.003 g
Acc Y	0.091 g	0.003 g
Acc Z	1.005 g	0.004 g
Gyro X	−0.007 dps	0.125 dps
Gyro Y	−0.040 dps	0.052 dps
Gyro Z	−0.017 dps	0.091 dps

Table 1: Sensor Noise Statistics from Stationary Test. Acc Z Reads $\approx 1\,\mathrm{g}$ Due to Gravity

Below are the plotted visualizations of Accelerometer and Gyroscope data as a function of time for the test cycles:

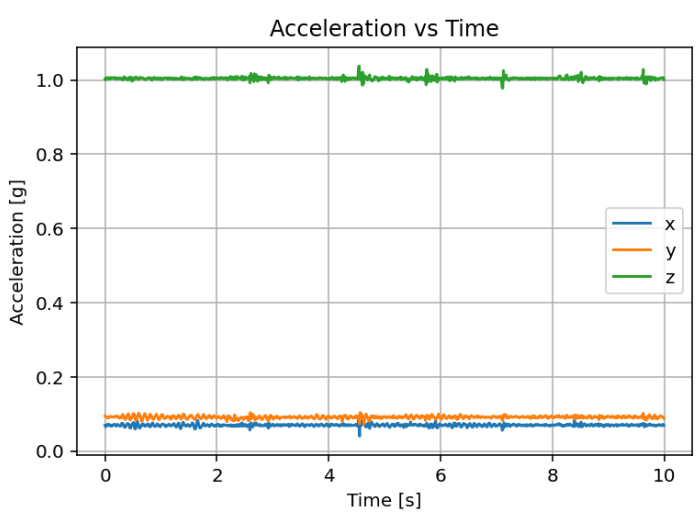


Figure 1: Acceleration vs. Time for Zero-G Measurements

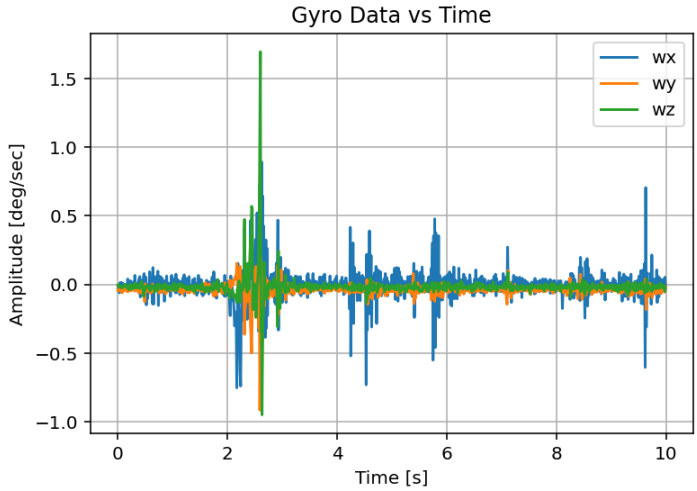


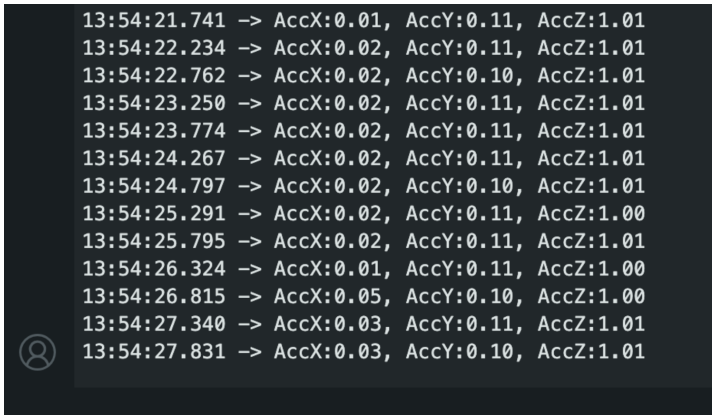
Figure 2: Gyroscope Angular Velocity vs. Time for Zero-G Measurements

The mean on Acc Z can be interpreted as reflecting the sensor orientation: the Z-axis was approximately aligned with gravity, which has an acceleration of 1 g. The relatively larger standard deviations in Gyro X and Gyro Y compared to the other channels could be due to residual low-frequency vibrations that had not fully decayed by the end of the recording window.

2 2g Test Results

The 2g calibration procedure (per LSM6DSO datasheet Section 4.6.1) was performed by orienting each axis repeatedly along $+g$ and $-g$ directions. A screen capture of the Arduino Serial Monitor output from the 2g test is shown in Figure 3. After initial cantilever beam testing, it was realized that the sensor range must be adjusted to ± 8 g to capture full oscillation ranges at the initial release. Below is an accelerometer test at the new range, along with example serial data responses recorded:

Figure 3: ± 8 g Accelerometer Bound Test on Arduino Serial Plotter
Acc X (Blue), Acc Y (Orange), Acc Z (Green)



```
13:54:21.741 -> AccX:0.01, AccY:0.11, AccZ:1.01
13:54:22.234 -> AccX:0.02, AccY:0.11, AccZ:1.01
13:54:22.762 -> AccX:0.02, AccY:0.10, AccZ:1.01
13:54:23.250 -> AccX:0.02, AccY:0.11, AccZ:1.01
13:54:23.774 -> AccX:0.02, AccY:0.11, AccZ:1.01
13:54:24.267 -> AccX:0.02, AccY:0.11, AccZ:1.01
13:54:24.797 -> AccX:0.02, AccY:0.10, AccZ:1.01
13:54:25.291 -> AccX:0.02, AccY:0.11, AccZ:1.00
13:54:25.795 -> AccX:0.02, AccY:0.11, AccZ:1.01
13:54:26.324 -> AccX:0.01, AccY:0.11, AccZ:1.00
13:54:26.815 -> AccX:0.05, AccY:0.10, AccZ:1.00
13:54:27.340 -> AccX:0.03, AccY:0.11, AccZ:1.01
13:54:27.831 -> AccX:0.03, AccY:0.10, AccZ:1.01
```

Figure 4: Accelerometer Sample Serial Log Response Data

3 Experimental Setup

3.1 Cantilever Beam Schematic

For the cantilever beam experiment, the accelerometer was taped directly over the free-end mass, with the sensor mounted normally to the top surface. The sensor is connected to the Arduino UNO, which can then be used to collect data with the Python DAQ programs to convert to .csv data.

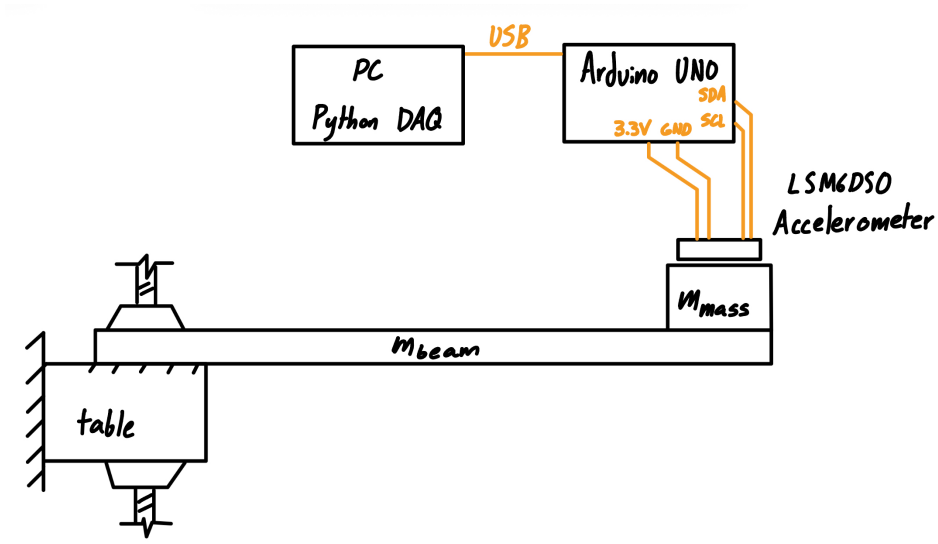


Figure 5: Schematic of the cantilever beam-mass experimental setup. The LSM6DSO accelerometer was mounted at the free end of the beam near the tip mass. The beam was clamped at one end, and the Arduino was connected to a PC via USB for data acquisition.

3.2 Setup Parameters

Parameter	Symbol	Value
Material Density (6061 Aluminum)	ρ	$2700\,\mathrm{kg}\cdot\mathrm{m}^2$
Young's Modulus	E	$68.9\times10^9\,\mathrm{N/m}^2$
Beam length (free)	L	$0.40\,\mathrm{m}$
Beam width	b	$0.0254\,\mathrm{m}$ (1 in)
Beam thickness	h	$0.003175\,\mathrm{m}$ (1/8 in)
Tip mass	m_{mass}	$0.060\,\mathrm{kg}$ (60 g)
Recording duration	T	$20\,\mathrm{s}$
Accelerometer range	—	$\pm8\,\mathrm{g}$
Gyroscope range	—	$\pm125\,\mathrm{dps}$

Table 2: Measured and Configured Parameters for the Cantilever Beam Experiment

4 Captured Acceleration Data

4.1 Raw Acceleration vs. Time

Figure 6 shows all three axes of raw accelerometer data over the 20-second recording. The Z-axis is the primary measured vertical and displays large-amplitude oscillatory decay. Comparatively, the amplitudes on the X and Y axes are much smaller throughout.

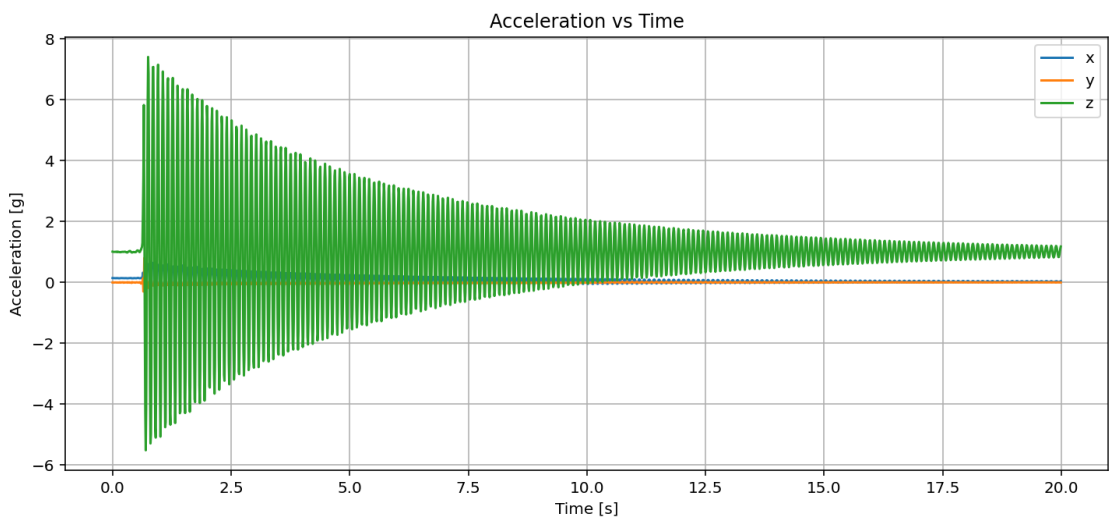


Figure 6: Raw Acceleration vs. Time for All 3 Axes (X, Y, Z)

4.2 Zero-Mean Acceleration with Detected Peaks

Peaks were automatically detected using `scipy.signal.find_peaks` with a prominence threshold of 0.01 g, capturing the local maxima of the decaying oscillation.

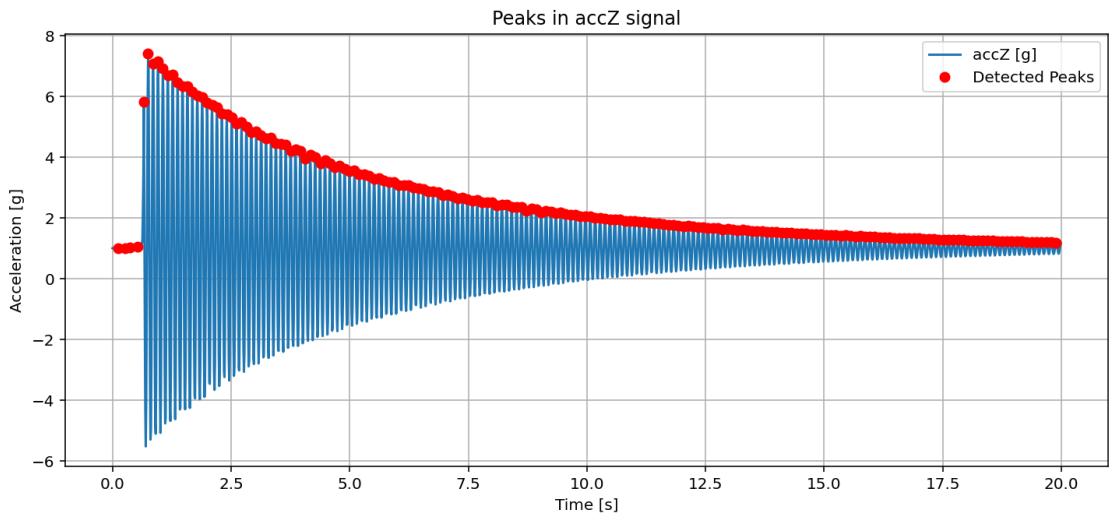


Figure 7: Zero-mean Acceleration (Z) vs. Time with Detected Peaks (Red)

4.3 Logarithmic Decrement and Linear Regression

The logarithmic decrement is defined as the natural logarithm of the ratio of successive peak amplitudes. For a linearly damped system, this is constant, so the estimate is obtained by fitting a line to $\ln(A_0/A_n)$ using the cycle number n. First, the 2nd-order response of the underdamped system must be represented as a differential equation of vertical position, x:

$$\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0 \tag{1}$$

Solving x as a function of t:

$$x(t) = e^{-\zeta\omega_n t}\left[\frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d}\sin(\omega_d t) + x_0\cos(\omega_d t)\right] \tag{2}$$

Which can alternatively be written as:

$$x(t) = A_0 e^{-\xi\omega_n t}\cos(\omega_d t - \phi) \tag{3}$$

Where:

$$\text{Amplitude: } A_0 = \sqrt{\left(\frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d}\right)^2 + x_0^2}$$

$$\text{Phase Angle: } \tan\phi = \frac{\dot{x}_0 + \zeta\omega_n x_0}{\omega_d x_0}$$

We can experimentally determine the amplitude ratio to determine damping ratio by taking the nth Amplitude over the next discrete amplitude value.

$$\frac{A_n}{A_{n+1}} = \frac{A_0 e^{-\zeta\omega_n t_n}}{A_0 e^{-\zeta\omega_n t_{n+1}}} = e^{\zeta\omega_n(t_{n+1}-t_n)}$$

$$T_d = t_{n+1} - t_n = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$\ln\left(\frac{A_n}{A_{n+1}}\right) = \ln\left(e^{\zeta\omega_n\left(\frac{2\pi}{\omega_n\sqrt{1-\zeta^2}}\right)}\right)$$

$$\ln\left(\frac{A_n}{A_{n+1}}\right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

Lastly, using logarithmic rules, we can take the product n from the amplitude ratio, which describes the change in amplitudes as a function of the number of cycles, n. The slope of this curve we can then simplify to a new variable, which will be denoted as beta (β).

$$\ln\left(\frac{A_0}{A_n}\right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}} \cdot n = \beta \cdot n \tag{4}$$

The β value represents the logarithmic decrement. The plot below shows this linear regression result.

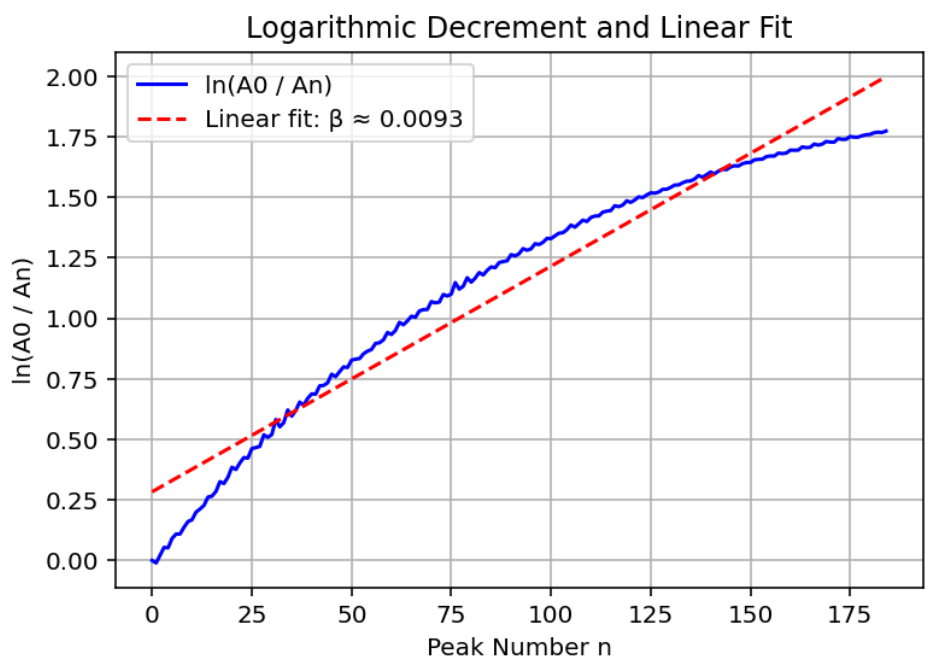


Figure 8: Logarithmic Decrement $\ln(A_0/A_n)$ vs. Cycle Number n with Linear Regression

5 Beam-Mass System Parameter Estimation

5.1 Structural Properties

The area moment of inertia (I) of the rectangular cross-section and the beam stiffness (k) of the cantilever beam are:

$$I = \frac{bh^3}{12} = \frac{(0.0254)(0.003175)^3}{12} = 6.775 \times 10^{-11}\ \text{m}^4 \tag{5}$$

$$k = \frac{3EI}{L^3} = \frac{3(68.9 \times 10^9)(6.775 \times 10^{-11})}{(0.40)^3} = 218.80\ \text{N/m} \tag{6}$$

The effective mass of the beam-mass system is approximated as the tip mass plus a correction for the distributed beam mass:

$$m_{\text{beam}} = \rho \cdot bhL = (2700)(0.0254)(0.003175)(0.40) = 0.0871 \text{ kg} \tag{7}$$

$$m_{\text{eff}} = m_{\text{mass}} + 0.23\, m_{\text{beam}} = 0.060 + 0.23(0.0871) = 0.0800 \text{ kg} \tag{8}$$

5.2 Dynamic Parameter Results

Parameter	**Symbol**	**Value**
Logarithmic decrement (slope)	β	0.0093
Damping ratio	ζ	1.4×10^{-5}
Damped period	T_d	0.1034 s
Damped natural frequency	ω_d	60.7517 rad/s (9.6689 Hz)
Undamped natural frequency (measured)	ω_n	60.7521 rad/s (9.6690 Hz)
Effective stiffness	k	218.80 N/m
Effective mass	m_{eff}	0.0800 kg
Theoretical $\omega_n = \sqrt{k/m_{\text{eff}}}$	$\omega_{n,\text{theory}}$	52.29 rad/s (8.32 Hz)

Table 3: Summary of Estimated Beam-Mass System Parameters

The damped and undamped natural frequencies are nearly identical, which is expected given the low damping ratio ($\zeta \approx 10^{-5}$). The measured ω_n is 16% higher than the predicted value of 52.29 rad/s. This likely reflects the uncertainty in free length L or h, which have cubic relationships on stiffness. The nominal length $L = 40$ cm was measured from the table edge to the tip mass; however, the fixed-end boundary condition may lie closer to the clamp jaw neutral axis. This overestimates L and therefore underestimates k and ω_n.

Back-solving from the measured $\omega_n = 60.75$ rad/s with the confirmed effective mass gives an effective free length of $L_{\text{eff}} \approx 36.2$ cm, roughly 3.8 cm shorter than the ruler-measure value, consistent with the physical size of the hand clamp jaw. The table below shows $\omega_{n,\text{theory}}$ over a conservative ± 3 cm range around the nominal length.

Length L	k **[N/m]**	$\omega_{n,\textbf{theory}}$ **[rad/s]**	$f_{n,\textbf{theory}}$ **[Hz]**
37 cm (min)	276.5	59.33	9.44
40 cm (nominal)	218.8	52.29	8.32
43 cm (max)	176.1	46.48	7.40

Table 4: Theoretical ω_n Bounds Under ± 3 cm in Beam Length. All Other Parameters Held Constant.

The resulting bounds on the predicted undamped natural frequency are:

$$46.48 \text{ rad/s } (7.40 \text{ Hz}) \;\leq\; \omega_{n,\text{theory}} \;\leq\; 59.33 \text{ rad/s } (9.44 \text{ Hz}) \tag{9}$$

The experimentally found value of $\omega_n = 60.75$ rad/s is slightly above the upper limit, as expected due to the boundary condition at the clamp being stiffer than that of an ideal fixed boundary. The reason for this is that the beam is resting on the table surface and is being compressed down onto it by the hand clamp, so that the table edge acts as a pivot point, increasing the boundary stiffness at the root beyond that predicted by theory. The boundary stiffness k is therefore greater than that predicted by theory, causing the predicted value of ω_n to be greater than that found experimentally.